

Traditional library operations rely heavily on manual record-keeping or fragmented spreadsheet tracking, which introduces significant operational inefficiencies, data redundancy, and administrative bottlenecks. As library inventories and member bases scale, managing book catalogs, tracking borrower histories, and enforcing return timelines through manual intervention becomes unsustainable.

The current system suffers from four critical operational flaws:

Inefficient Cataloguing and Search: Borrowers and staff waste time manually searching physical registers or shelves to locate books, with no real-time visibility into whether a book is currently available, reserved, or misplaced.

Tracking and Security Vulnerabilities: Manual logs fail to accurately track the moving chain of custody for books, leading to unrecorded losses, unauthorized borrowing, and discrepancies in inventory audits.

Prone to Human Error: Manual calculation of overdue days and fine structures often results in mathematical errors, inconsistent policy enforcement, and disputes between members and administration.

Lack of Data-Driven Insights: Administrative staff cannot easily compile borrowing trends, identify underutilized inventory, or track active user demographics due to the absence of centralized data analytics.